distance between clusters C_1 and C_2, but it's not small enough to keep clusters C_3 and C_4 separated.

If we try all possible values of epsilon and keep track of the resulting clustering, we can analyze how the partitioning evolves: the best way to perform this analysis is through a dendrogram, a tree-like structure shown in figure 12.22, where – for the sake of clarity - the x axis only shows as lowest-level entries the 8 clusters (plus some noise) in figure 12.17, while it would normally have one entry per point in the dataset. Notice how the clusters and noise points are ordered along the x axis of the dendrogram: all points in this plot must follow the same order as in the reachability plot (which, in turn, is the same order points are processed by OPTICS).